Dresden Nuclear Power Station
6500 North Dresden Road
Morris, IL 60450

October 23, 2025

10 CFR 50.73

SVPLTR 25-0065

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Dresden Nuclear Power Station, Unit 2
Renewed Facility Operating License No. DPR-19
NRC Docket No. 50-237

Subject: Licensee Event Report 237/2024-001-01, Containment Cooling Service Water Valve Failure Resulted in a Condition Prohibited by Technical Specifications and Loss of Safety Function

Enclosed is a revision to Licensee Event Report 237/2024-001-01, Containment Cooling Service Water Valve Failure Resulted in a Condition Prohibited by Technical Specifications and Loss of Safety Function. This report describes an event being reported in accordance with 10 CFR 50.73(a)(2)(i)(B), 10 CFR 50.73(a)(2)(v)(B) and 10 CFR 50.73(a)(2)(v)(D) for a Condition Prohibited by Technical Specifications and Condition that Could Have Prevented Fulfillment of a Safety Function.

There are no regulatory commitments contained in this submittal.

Should you have any questions concerning this letter, please contact Mr. Daniel J. Murphy, Regulatory Assurance Manager, at (779) 231-7443.

Respectfully,

Hardik Patel

Hardik Patel
Site Vice President
Dresden Nuclear Power Station

Enclosure: Licensee Event Report 237/2024-001-01

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Dresden Nuclear Power Station, Unit 2	☒ 050 ☐ 052	00237	1 OF 5

4. Title: Containment Cooling Service Water Valve Failure Resulted in a Condition Prohibited by Technical Specifications and Loss of Safety Function

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
04	30	2024	2024	001	01	10	23	2025	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1 – Power Operation	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20		10 CFR Part 50			
	☐ 20.2203(a)(2)(vi)		☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☒ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Daniel J. Murphy, Manager, Site Regulatory Assurance	(779) 231-7443

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	BI	FD	B135	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 4/30/2024 at 10:07 CST when starting a Containment Cooling Service Water (CCSW) Pump, the Dresden Unit 2 Containment Cooling Heat Exchanger Division I Tube Side Discharge Valve failed to open. The safety related function of the valve is to control cooling water flow through the heat exchanger tubes to remove heat from containment. The condition was corrected at 03:10 on 5/1/2024. The station determined the Division I CCSW system was inoperable for 28.5 days, therefore, an Operation or Condition Prohibited by Technical Specifications existed for greater than the Limiting Condition of Operation (LCO) time of 7 days. During this period, surveillance testing performed on 4/3/2024 resulted in the Division II subsystem being inoperable for 3 hours and 39 minutes resulting in an Event or Condition that Could Have Prevented Fulfillment of a Safety Function. The root cause of the event was a cracked yoke weld that allowed the actuator to flex away from the valve body. This resulted in overtravel of the valve position potentiometer during valve closure, causing an open permissive in the valve's control logic. This was corrected by a weld repair, potentiometer and valve stroke length adjustments. This report is being submitted under 10 CFR 50.73(a)(2)(i)(B), 10 CFR 50.73(a)(2)(v)(B) and 10 CFR 50.73(a)(2)(v)(D).

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric – Boiling Water Reactor, 2957 megawatts thermal rated core power

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

CONDITIONS PRIOR TO EVENT

Unit: 2	Event Date: April 30, 2024	Event Time: 10:07 CST
Reactor Mode: 1	Mode Name: Power Operation	Power Level: 100 percent

No structures, systems, or components were inoperable at the time of the event that contributed to the event.

A. DESCRIPTION OF EVENT

On 4/30/2024 at 10:07 CST, with Dresden Unit 2 at 100 percent power, while performing operations to support Suppression Pool [BO] level control, no CCSW [BI] flow was observed on the instrumentation in the Main Control Room (MCR) following system start. When 2A CCSW [BI] Pump was started per Dresden Operating Procedures, the Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] did not automatically open (dual indication) as expected. The system is designed to remove energy from the suppression pool [NH] by rejecting heat to the CCSW [BI] system. Suppression pool cooling [BO] or containment [NH] sprays may be initiated following a Loss of Coolant Accident (LOCA) using a portion of the non-selected Low Pressure Coolant Injection (LPCI) [BO] loop, which is not required to perform a LPCI [BO] function. When CCSW [BI] is initiated, the Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] is designed to open based on demand from a Differential Pressure (DP) Controller [PDC].

A past operability review was completed on 6/20/2024 and determined that the Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] was incapable of performing its function after the valve [PDCV] was opened and closed for Post Maintenance Testing (PMT) at 14:24 CST on 4/2/2024. This was not discovered until the past operability was reviewed for reporting criteria on 11/27/2024 and determined to be reportable under 10 CFR 50.73.

When the Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] failed to open on 4/30/2024, Technical Specification (TS) 3.7.1 Conditions A and C were entered for Division 1 CCSW [BI] being declared inoperable. Division 1 Suppression Pool Cooling [BO], Suppression Pool [NH] Spray and Drywell [NH] Spray were also INOPERABLE (TS 3.6.2.3, 3.6.2.4 and 3.6.2.6 respectively). The Limiting Condition of Operation (LCO) time for one subsystem inoperable is 7 days, therefore an Operation or Condition Prohibited by TS existed from 4/10/2024 02:24 CST to 5/1/2024 03:10 CST. This is being reported under 10 CFR 50.73(a)(2)(i)(B).

Additionally, from 00:42 to 04:21 CST on 4/3/2024, the opposite Division II CCSW [BI], Suppression Pool Cooling [BO], Suppression Pool [NH] Spray and Drywell [NH] Spray were inoperable due to surveillance testing on Unit 2 LPCI [BO] Initiation circuitry. This also resulted in an Event or Condition that Could Have

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Prevented Fulfillment of a Safety Function and is being reported under 10 CFR 50.73(a)(2)(v)(B) and 10 CFR 50.73(a)(2)(v)(D).

B. CAUSE OF EVENT

The Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] failed to open from the MCR due to potentiometer [FD] over-range feedback causing the downstream relay [RLY] to remain open. Without this open permissive being made up, the valve [PDCV] was unable to be operated from the MCR. The direct cause of the valve [PDCV] failure to operate was the position feedback potentiometer [FD] over-ranging, thus preventing the Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] from responding to demand from the controller [PDC] as required. The root cause of the failures was a cracked weld between the actuator mounting plate and the top of the valve [PDCV] yoke. This allowed the actuator to flex away from the valve [PDCV] body, effectively increasing the stroke length. This degraded condition resulted in over-ranging of the potentiometer [FD]. Contributing, this weld was performed in 1977 following a yoke replacement/repair, and is the only one of the four CCSW Heat Exchanger Discharge Valve [PDCV] yokes with any evidence of work having been performed. A bad weld was likely introduced in 1977 and was subject to low load cyclic fatigue which contributed to the slow propagation of a crack along the heat affected zone on the actuator mounting plate side of the weld. The valve [PDCV] stroke length was found to be using nearly the full band of the potentiometer [FD] and required little additional movement for overtravel into the dead-band region when the valve [PDCV] was taken to the full closed position. The design of the 2(3)-1501-3A(B) valves logic is not fault tolerant, where position feedback potentiometers feed into the control logic and, if over-ranged, will cause the valve to fail to respond to demand from the controller.

The valve [PDCV] had previously been cycled open and closed at 14:24 on 4/2/24 as part of PMT. The valve [PDCV] was not operated again until the failure was identified at 10:07 on 4/30/24. On 4/2/24 the potentiometer [FD] would have over-traveled and caused a failure to open at the next open demand from the Differential Pressure (DP) Controller [PDC] within the MCR. This condition would not have been apparent until the next attempt to operate the valve [PDCV] via the MCR. With the failure mechanism being associated with the potentiometer [FD] over-travelling when the valve [PDCV] is taken to full close, there is reasonable assurance the valve [PDCV] would have performed its function prior to 4/2/24.

C. SAFETY ANALYSIS

The Dresden Updated Final Safety Analysis Report (UFSAR) long-term Design Basis Accident-LOCA (DBA-LOCA) analysis assumes one Containment Cooling loop [BO] with one heat exchanger [HX] is available for containment [NH] cooling, being manually initiated at 10 minutes. Assuming the suppression pool [NH] and CCSW [BI] initial temperature is at TS limits of 95 degrees Fahrenheit, the peak suppression pool [NH] temperature is 196 degrees Fahrenheit. The heat removal capability of one subsystem is sufficient to meet the overall DBA cooling requirement for LOCAs and transient events. Additional margin to containment limits is provided by nominal plant parameters.

For 3 hours and 39 minutes, while in Modes 1, 2 and 3, both divisions of Suppression Pool Cooling [BO], Suppression Pool [NH] Spray and Drywell [NH] Spray were inoperable. Division I was inoperable due to the U2 Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV], preventing CCSW [BI] flow. Division II was

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inoperable due to LPCI [BO] pump breakers [BKR] being racked to test. In this condition, LPCI [BO] flow from Division I was capable of providing flow to containment [NH] sprays. Without cooling from CCSW [BI], this spray flow would cool noncondensible gases, condense steam, and assist the natural convection and diffusion mixing of hydrogen and oxygen in the containment following a postulated LOCA. The flow would also continue to provide scrubbing by the containment [NH] sprays to reduce the activity of elemental iodine and aerosols released from the core.

With two subsystems inoperable, an 8 hour Completion Time to restore one is based on the loss of function, and is considered acceptable due to the low probability of a DBA and the potential avoidance of a plant shutdown transient that could result in the need for the subsystems to operate. Operators were able to cycle the U2 Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] locally using operating procedures. Additionally, operators had been briefed, were standing by, and at least one LPCI [BO] pump on Division II would have been able to be restored in approximately 10 minutes, with the second LPCI [BO] pump being restored two to three minutes later. During a LOCA (assuming conservative 10 CFR 50 Appendix K model numbers), peak cladding temperature reaches 2136 degrees Fahrenheit, which will result in some fuel failure and an increase in radiation in the areas required to restore at least one division. During nominal plant operational parameters, fuel cladding remains below 1500 degrees Fahrenheit following a DBA-LOCA, and therefore, based on Engineering Judgement, the radiation and temperatures in these areas is expected at levels commensurate with normal operating conditions. During the 3 hours and 39 minutes, suppression pool [NH] temperature was a nominal 79 degrees Fahrenheit, CCSW [BI] was a nominal 77 degrees Fahrenheit, providing additional margin to limits.

A. CORRECTIVE ACTIONS

Corrective actions adjusted the potentiometer [FD] and the stroke length for the Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV]. The cracked weld was repaired and extent of condition inspections were performed on the other three Heat Exchanger [HX] Tube Side Discharge Valves [PDCV]. Compensatory actions have been put in place to verify/validate the logic following each operation until modifications have been completed and updates were made to the database sheets with the limitations associated with the potentiometer [FD].

Long-term corrective actions are to perform:

Modifications to the Heat Exchanger [HX] Tube Side Discharge Valves [PDCV] logic to remove/modify position feedback so as to not prevent the valve from performing its function; replacement of the 'A' Tube Side Discharge Valve [PDCV] yoke; and adjustments of the other division and Dresden Unit 3 Heat Exchanger [HX] Tube Side Discharge Valves [PDCV]. However, these three have been verified to have margin on the potentiometers [FD] during valve [PDCV] cycling and have no history of yoke repairs. The modification will also support prevention of re-closure due to torque switch relaxation. Additional correctives include replacement of the yoke and mounting plate on the 2-1501-3A.

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B. PREVIOUS OCCURRENCES

A review of NRC LERs and the internal Corrective Action Program (CAP) was performed ranging back to 1971.

A yoke repair/replacement was performed on the Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] which included performing the weld that degraded in 1977, and is the only one of the four CCSW Heat Exchanger Discharge Valve [PDCV] yokes with evidence of work being performed.
The Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] failed to open due to the potentiometer on 06/03/1977, unknown causes on 12/16/1977, potentiometer on 11/21/1983, and 02/29/1996.
On 11/14/2023, the Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] failed to operate during a CCSW [BI] Loop Flow Verification surveillance, while returning the system to service. The direct cause of the valve [PDCV] failure to operate on was the position feedback potentiometer [FD] over-ranging, thus preventing the Heat Exchanger [HX] 'A' Tube Side Discharge Valve [PDCV] from responding to demand from the controller [PDC] as required. The root cause of the failures was the cracked weld between the actuator mounting plate and the top of the valve [PDCV] yoke. This event did not result in result in a reportable condition.

C. COMPONENT FAILURE DATA

System: BI
Component: FD
Manufacturer: B135
Nomenclature: Single Turn Potentiometer
Model Number: 5611R1KL.5